

PARTICULAR SPECIFICATION P2E

APPENDIX P2E – Schedule of Submissions

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P2E.1 SCHEDULE OF SUBMISSIONS

- (a) Contractor shall comply with the following schedule for the submission of detailed information and documentation for the S.O.'s review. Models, drawings, calculations and other required documentation to be submitted shall be in accordance with the requirements of the General Specification G1.2 Information Management and P2.15 Information Management
- (b) The submission of detailed information and documentation for this Contract is described in Table P2E.1. The requirements in here complement the general requirements in the General Specifications and the particular requirements in the other Particular Specifications. Where there is a contradiction between the details in different sections of the Specifications, the Contractor shall seek clarification in writing from the S.O. before proceeding with provision of the affected items.

Table P2E.1 Schedule of Submissions

Submission/ review stage	Documents to be submitted/reviewed
Basis of Design Report (BODR)	The BODR shall be arranged in three volumes, as follows: Volume 1 Description of the Works. Volume 2 Process Functional Descriptions. Volume 3 Model and Drawings Calculations for each process unit and for each item of equipment in the process. Calculations shall be arranged in an Appendix to Volume 1 and shall include, but not be limited to: a) Mass and heat balance calculations and flow diagrams; b) Hydraulic and process design; c) Range of flows, pressure, drawn power and motor ratings for pumps and other mechanical equipment; d) Any other major item of equipment to justify the size, pressure rating, duty point, motor rating, etc, of the equipment; and e) Safety Plan.
50% and 85% Design submission	1. 6D BIM Model including: a. Plans and elevations of all buildings and structures b. Piling plans c. Equipment layouts; d. Plinth details (including loadings); and e. Electrical/ICA Duct banks and manholes; 2. P&ID drawings; 3. Process Control Narratives (PCN); 4. Functional Design Specifications (FDS); 5. Instrument list; 6. Instrument hook-up drawings;

Submission/ review stage	Documents to be submitted/reviewed
	7. Detailed control system network and architecture drawings, hook-up diagram showing all equipment model numbers, network connections and peripherals; 8. Isolation and Control Philosophy; 9. Pump schedules and data sheets; 10. Cable schedules; 11. Interface and tie in details; 12. Input/ Output (I/O) list; 13. Equipment list; 14. Motor/Drive list; 15. Valve list; 16. Penstock/Stop Log List; 17. Pipe/Materials list; 18. Lifting plan; 19. Finalised and complete electrical Single Line Diagrams (SLD); 20. Power factor correction; 21. Capacitor rating selection; 22. LVSG, MCC and MDB Single Line Diagram; 23. CCTV and Access Control layouts; 24. Building services equipment layouts; 25. Lighting and small power layouts; 26. General area lighting layout; 27. Schematic drawings for plant water supply system and any other ancillary systems; 28. Drainage layout.
Design Calculations	1. Electrical load list; 2. Transformer sizing calculations; 3. Electrical fault level calculations; 4. Electrical earthing calculations; 5. Lightning protection calculation, risk assessment; 6. Head load and ventilation calculations; 7. Illumination calculations; 8. Substructure and superstructure design calculations; 9. Temporary works design calculations; 10. Steelwork design calculations; 11. Piping and valve sizing calculations;

Submission/ review stage	Documents to be submitted/reviewed
	12. All Equipment including Pumps, Heat Exchanger, future odour control facility Sizing Calculations;
HAZOP workshop	1. Complete HAZOP documentation, including HAZOP report, action list and P&ID's
ALMOP workshop	1. Complete ALMOP documentation, including ALMOP report, action list
Final design submissions	1. Amended BIM model and P&IDs resulting from HAZOP and ALMOP workshops; 2. Detailed pipework and valving layouts; 3. Detailed control system architecture and network drawings; 4. Transformer, LVSG, MCC and MDB general arrangement drawings; 5. Motor starter schematic drawings; 6. Motor detailed drawings; 7. Final equipment schedules; 8. Valve schedules; 9. Cable schedules; 10. Cable sizing and voltage drop calculations; 11. Cable tray sizing with calculations; 12. Software functional design specification (FDS); 13. Instrument data sheets; 14. Instrument hook-up drawings; 15. I/O list; 16. Finalised load list; 17. Finalised LVSG, MCC and MDB Single Line Diagrams; 18. Power factor correction calculations; 19. Power control and instrumentation cable schedules; 20. Cable routing layout, cable fill calculations, raceway sizing and routing layout; 21. Motor control schematics; 22. Final electrical equipment schedule; 23. MCC, LV Switchboard internal arrangement, section details, busbar arrangement; 24. AC approved and PE certified superstructure and substructure calculations and drawings; 25. Authorities clearances.

Submission/ review stage	Documents to be submitted/reviewed
Final design review	1. Submission of all final design documentation
Amended final designs submissions	1. Amended final designs; 2. Submission of all amended final design submission including a Final design report; 3. Submission of all amended final design for Development Control submission to URA; 4. Submission of all amended final design for Building Plan submission to the relevant technical departments and BCA; 5. Submission of all amended final design for Structural Plan submission to the relevant technical departments and BCA; 6. All as-built plans for submission; 7. Application Approval of Temporary Occupation Permit.

- (c) BIM models shall show the layout of Works in this Contract. As a minimum, the model shall include the following:
- (i) A plan view of each floor where equipment is housed, and elevation/section views from the side and from the front;
 - (ii) All primary units of mechanical and electrical equipment, including, the outline layout of electrical rooms;
 - (iii) All maintenance and access platforms;
 - (iv) All equipment lifting and maintenance appurtenances including cranes, monorails, fixed lifting beams, and similar appurtenances;
 - (v) Process piping and ductwork with diameter or maximum cross-section dimension of greater than or equal to 80 mm nominal bore. Main process piping which are less than 80 mm nominal bore shall be shown as well;
 - (vi) Locations for all interfaces with structural, mechanical, and electrical equipment supplied under other contracts; and
 - (vii) Routing of piping, ductwork and major electrical and ICA cabling.
- (d) The heat and mass balance (HMB) for Digestion Pre-treatment must include, at a minimum, the following:
- (i) Dry solids mass rates (kgDS/d), total wet mass (dry solids and moisture) rates (kg/d), percent dry solids concentrations, temperatures (degrees Celsius), and pressures (kPa) for each process, recycle, discharge and exhaust process gas stream;

- (ii) Heating and fuel requirements (steam flows), including temperatures (degrees Celsius) and pressures (kPa); and
 - (iii) Cooling water requirements, itemised for each point of usage within the system, including flow rates, temperature and pressure of cooling water supply and cooling water discharge streams.
- (e) The heat and mass balance (HMB) for Sidestream Nitrogen Removal facility must include, at a minimum, the following:
 - (i) Aeration flowrates requirement to each Deammonification reactor (Nm^3/h) including supply temperatures ($^{\circ}\text{C}$) and pressures (kPa);
 - (ii) The ammonia and nitrogen loadings of the inlet and outlet streams shall be included in the mass balance;
 - (iii) The solids loading before and after Equalisation/Solids removal tank, sludge wasting flowrates;
 - (iv) Process (N_2) gas production flowrate from Deammonification reactor, and any other by-product flowrates; and
 - (v) Chemical dosing flowrates (if required).
- (f) The detailed schematic process flow diagram and process control narrative of the system shall fully illustrate interconnection of equipment, location of critical monitoring and control instrumentation and describe the functionality of the process. The streams in the process flow diagram must be cross referenced to the heat and mass balance summary, noted above, to completely describe the process.
- (g) The Process Control Narratives (PCNs) shall describe the goals and intent of each unit process. It shall provide a description of the process, including key operating parameters, control schemes, and interfaces to the plant monitoring and control system. The description shall include the purpose of, and process reactions expected from, each control device for all modes of operation, including reactions to fault conditions. The description shall also include all manual actions required of the plant operator. Automatic operations shall be clarified as to whether they are discrete (ON/OFF or step) type actions and or continuous (proportional or gradual) type actions. The PCNs shall take reference to the Digestion Pre-treatment and Sidestream Nitrogen Removal PCN in Particular Specification Appendix P6A and Standard Control Functions PCN provided in Particular Specification Appendix P6B.

- (h) The software functional design specification (FDS) shall meet the following requirements:
 - (i) The FDS shall comprise an overall description of the Works, its functioning and control, and a detailed description of each section of the control system covering modes of operation, manual overrides, set point and parameter selection and adjustment. The detailed description shall include a step by step control description which defines the function of each piece of equipment and each control action and interlock, including details of the programme in each programmable item. The format of the FDS shall be logically grouped into process areas with sections as specified in the following clauses.
 - (ii) A general sub-section. This shall include a brief introduction of the process area, a list of major process plant and quantity of each type;
 - (iii) Any operational details of the control system which influence plant operations, such as power failure response, communication failure response, and automatic shut-down and start-up of the system;
 - (iv) Alarm sub-section. List of all alarms related to the process area;
 - (v) A sub-section of control flowcharts or control sequence descriptions for control loops. This shall contain both auto and manual features;
 - (vi) Fault sub-section. A list of all faults and trips that are expected to occur with a description for each;
 - (vii) Interlock table. This shall include both process and hardwire interlocks to quickly identify source of trip or fault;
 - (viii) The format shall be chosen to satisfy the requirements of the software design documentation;
 - (ix) The FDS shall describe the "fail safe" features incorporated into the design. This would, for example, describe what actions would be taken by the control system in the event of failure of a plant item or system, or loss of an input signal affecting a control loop or process sequence;
 - (x) The FDS shall describe control actions taken and monitoring functions which remain available during a power failure, and any automatic controls or sequencing which takes place during system start up and shutdown;
 - (xi) The FDS shall be presented in a clear and precise manner and shall include figures or drawings where appropriate;
 - (xii) The Contractor shall submit and obtain the S.O.'s approval of the FDS before starting the detailed control system design.

- (i) The list of equipment, to include as a minimum, primary equipment design criteria, proposed manufacturer's name and model number, and shipping and operating weights. Append and cross reference equipment catalogue descriptions to the equipment list, where applicable.
- (j) The schematic and calculations of utility requirements shall be those required for total system operation, to include steam, energy, plant water (G2D) and cooling water (G2D). This summary shall be based on the same conditions as the heat and mass balances required above. Utility requirements shall be itemised for each equipment component and shall list peak volumetric flow rate (L/s), maximum and average demand (cubic metres per hour), pressure (kPa).
- (k) The safety plan shall address equipment and features incorporated into the design, control attributes, and procedures to be followed for start-up, shutdown, normal operation, maintenance and emergency shutdown conditions including power failure, to assure a safe working environment for the Works.
- (l) The lifting plan, including narrative (where necessary) and drawings, of equipment hoisting and maintenance access to be provided. This description shall include a summary plan for the maintenance of, and removal/replacement of, all items of major equipment.
- (m) A narrative description of recommended provisions for emergency and standby power shall be provided and shall be co-ordinated with the safety plan. Emergency and standby power if necessary, for the safe operation and maintenance of the Works shall be provided and included in the Contract Sum.

END OF SECTION